

Response Letter with Annotations

Xu, Y. (2024). Formation mechanisms of intra-organizational membership overlap: a longitudinal network analysis of membership data from the International Communication Association. *Journal of Communication*, 74(4), 333–345. <https://doi.org/10.1093/joc/jqae021>

Annotations

1. Purposes

- a. Motivation (one-sentence summary): Explain how and why ICA divisions and interest groups come to share members over time.
- b. Specific Rationale (several bullet points):
 - The authors started with a basic phenomenon: shared members (“membership overlap”), when many people belong to more than one group, which creates links between those groups. In this case, they are focusing on ICA, International Communication Association, divisions and interest groups.
 - They treated those links as a network that changes over time, so each subunit is a node. An connection between groups (which they called as edges) exists if two subunits share more members than expected. Looking year-by-year turns this into a longitudinal network of overlap.
 - Then, they asked: which mechanisms could generate this evolving network? The paper proposed four concrete hypothesis and tested them
 - The middle-range niche theory: By mapping these concrete mechanisms onto ecological concepts such as niche overlap, the study demonstrates how the abstract ecology-of-affiliation theory operates within the context of a real professional association.
 - The authors used an specific statistical tool to test the hypothesis, which is the temporal exponential random graph models (TERGMs) that simulates many possible networks and check which combination of mechanisms best reproduces the real overlap patterns across nine years
 - The results: there is no single dominant subunit, overlap grows mostly via triadic closure, small subunits depend heavily on overlap (especially with large ones), and co-sponsored sessions reliably foster dual memberships.
- c. Research Questions and/or Hypotheses
 - Identify IV, DV, mediator, moderator
 1. **DV:** Membership overlap network (edge presence): For each pair of ICA subunits in year t , the outcome is whether there is an edge (1 when overlap is significantly above threshold, 0 means no edge)

2. **IVs:** In order to understand why a connection (edge) appears between two ICA groups the authors asked what increases the chances of two groups share members. To answer that, they created some IV that represent theoretical mechanisms.
 - a. Network concentration (gwdegree) – RQ1
 - b. Triadic closure (triangle) – H1
 - c. Subunit size (nodecov subunit size) – H2
 - d. Size difference (absdiff subunit size) – H3
 - e. Network multiplexity / affinity (edgescov affinity) – H4 / RQ2
 - f. Affinity over time (timecov affinity x time) – RQ2
3. **Identify specific relationships among variables:** In this TERGM, all variables relate to the same DV, whether a membership-overlap edge between two ICA subunits is present in year t . This means that all IVs tried to explain the presence of an edge, but through different theoretical mechanisms.

2. Method

a. Research Design

- Longitudinal network analysis of 9 yearly snapshots (2007–2015) of ICA members' affiliations with 24 subunits.
- The data come from ICA membership records.
- The authors first constructed annual two-mode matrices (members x subunits) indicating whether a member was affiliated with a division or interest group in a given year. These matrices were then converted into one-mode networks of subunits, where an edge represents significant membership overlap between two subunits.
 1. Each cell was coded as 1 if an ICA member was affiliated with a specific division or interest group in year t , and 0 if not.

b. Sample

- All ICA members with any subunit affiliation in 2007–2015 were included, no sampling was done
- Units of analysis in the network: 24 ICA divisions/interest groups founded no later than 2007.
- This is effectively a complete network of these 24 subunits over nine years, not a representative sample.

c. Measures

- **Membership overlap (DV):** It is a binary edge between two ICA subunits in year t . It takes the value 1 when the number of shared members between two subunits is significantly above a statistical threshold, and 0 otherwise. This indicates whether a membership-overlap relationship exists between two subunits.
- **Network concentration (gwdegree):** It captures the distribution of edges across the network and tests whether overlap ties tend to concentrate on already well-connected subunits.

- **Triadic closure (triangle):** It counts closed triads in the network. It captures whether two subunits that both share members with a third subunit are more likely to form a direct overlap tie.
- **Subunit size (nodecov subunit size):** It measures the number of members in each ICA subunit, testing whether the size of a subunit influences the likelihood of forming overlap ties.
- **Size difference (absdiff subunit size):** It calculates the absolute difference in size between two subunits, capturing whether subunits of different sizes are more likely to share members.
- **Affinity (edgecov affinity):** It measures whether two subunits co-sponsor conference sessions, representing an affinity or collaboration tie between them.
- **Affinity over time (timecov affinity x time):** It captures the interaction between affinity and time, testing whether the effect of co-sponsorship on membership overlap changes across the years.

d. Analysis

- To test the hypotheses, the authors used **Temporal Exponential Random Graph Models (TERGMs)**. TERGMs are statistical models used to analyze networks that change over time. They estimate the probability that an edge (connection) exists between two nodes in the network. In this study, the model was used to estimate whether there was a membership-overlap edge between two ICA subunits in year t , based on several independent variables (IVs). In other words, the TERGM tests which factors increase or decrease the probability of membership overlap between subunits. The models were estimated using the **btergm package in R**, which allows researchers to run TERGM models, analyze longitudinal network data, estimate coefficients, and calculate standard errors and confidence intervals. To estimate the model parameters, the authors used **MCMC-MLE (Markov Chain Monte Carlo Maximum Likelihood Estimation)**. This is a statistical estimation method that calculates the most likely values for the model parameters. The method generates many simulated networks, compares them with the observed network, and adjusts the parameters until the simulated networks resemble the real network. This approach is more computationally demanding but usually produces more accurate estimates.

3. Results and Conclusions

a. Answers to RQs and Hypotheses

- **RQ1 – Network concentration:** RQ1 examined whether membership overlap becomes concentrated around a few highly connected subunits. The results show that the *gwdegree* term is not statistically significant, indicating that overlap ties do not systematically concentrate around already well-connected subunits. Therefore, there is no evidence of network concentration.
- **H1 – Triadic closure is supported:** H1 predicted that triadic closure would increase the likelihood of membership overlap between subunits. The *triangle* term is positive and highly significant, indicating that

subunits that share members with a common third subunit are more likely to form direct overlap ties. Thus, H1 is supported.

- H2 – Subunit size is supported: H2 predicted that the size of a subunit influences the likelihood of membership overlap. The coefficient for subunit size (nodecov) is negative and significant, indicating that larger subunits are less likely to form overlap ties. Therefore, H2 is supported.
- H3 – Size difference between subunits is partial supported: H3 predicted that differences in subunit size affect the probability of overlap. The absdiff size coefficient is positive and marginally significant, suggesting that overlap ties are somewhat more likely between subunits of different sizes. Thus, H3 receives partial support.
- H4 – Network multiplexity (affinity) is supported: H4 predicted that co-sponsorship ties (affinity) increase the likelihood of membership overlap. The affinity coefficient is positive and significant, indicating that subunits that co-sponsor conference sessions are more likely to share members. Therefore, H4 is supported.
- RQ2 – Affinity over time: RQ2 examined whether the effect of co-sponsorship on membership overlap changes over time. The interaction term (Affinity x Time) is not statistically significant, indicating that the relationship between co-sponsorship and membership overlap remains stable over time.

b. Include Key Figures/Tables

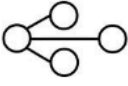
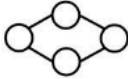
- Table 1: Overview of the membership overlap network over time

Table 1. Descriptive statistics of the observed network of membership overlap, 2007 to 2015

Time point	Number of present edges	Network density ^a	Number of closed triads ^b	Transitivity ^c	Average degree of separation ^d	Modularity ^e
2007	131	.47	281	.60	1.55	.21
2008	160	.58	484	.69	1.43	.14
2009	149	.54	389	.64	1.47	.15
2010	155	.56	441	.66	1.45	.15
2011	147	.53	400	.66	1.48	.14
2012	163	.59	510	.69	1.41	.12
2013	166	.60	540	.70	1.40	.13
2014	178	.64	635	.73	1.36	.09
2015	185	.67	692	.74	1.33	.08

- Table 2: Results of the TERGM predicting membership overlap

Table 2. Results of the TERGM predicting membership overlap

	Model 1 (baseline)		Model 2 (full)	
	Estimate	SE	Estimate	SE
Terms for structural configurations				
Edges	1.297***	0.211	1.387	1.475
				
2-stars			-0.233*	0.103
				
3-stars			0.010	0.007
				
4-cycle			-0.047***	0.006
				
RQ ₁ : Gwdegree			1.755	1.088
H ₁ : Triangle			1.056***	0.053
Terms for node attributes				
Subunit age (nodecov)	-0.017***	0.003	-0.002	0.002
Subunit age (absdiff)	-0.026***	0.004	-0.013***	0.003
Subunit status (nodecov)	-0.333	0.183	-0.165	0.157
Subunit status (nodematch)	0.420*	0.195	0.429*	0.173
H ₂ : Subunit size/100 (nodecov)			-0.107***	0.020
H ₃ : Subunit size/100 (absdiff)			0.044*	0.022
Terms for edge attributes				
Time (timecov)	0.117***	0.017	0.128***	0.031
H ₄ : Affinity (edgescov)			1.114***	0.316
RQ ₂ : Affinity × Time (timecov)			0.022	0.061
Area under the precision-recall curve	.675		.763	

Note.

* $p < .05$, *** $p < .001$.

- Figure 2: Network of membership overlap among ICA subunits

Formation mechanisms of membership overlap

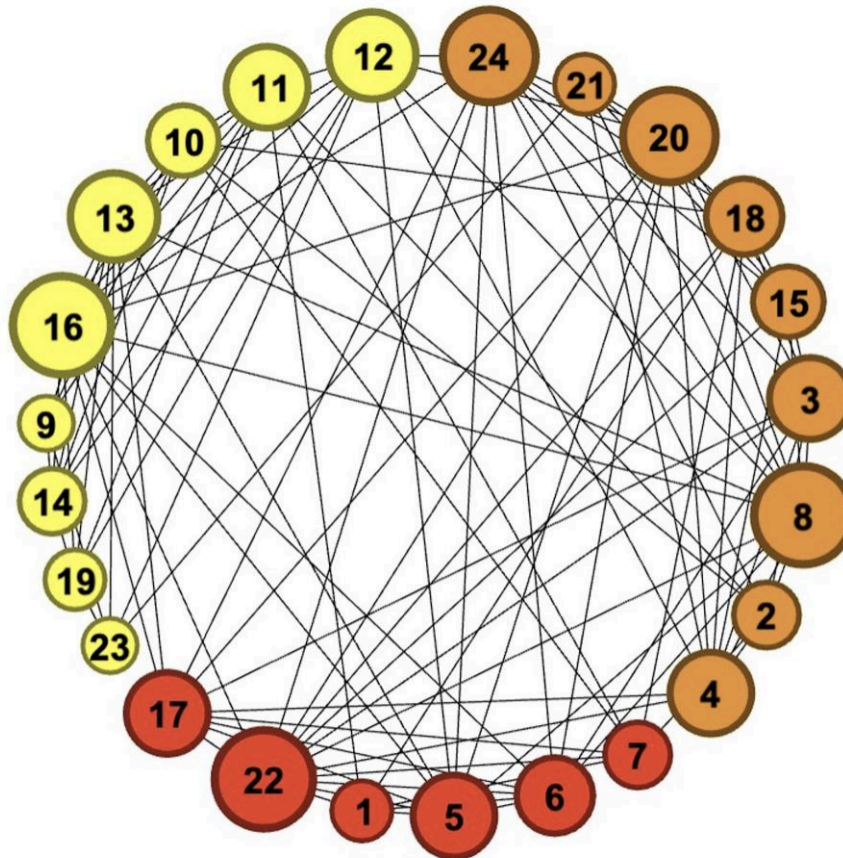


Figure 2. A network visualization of the structure of membership overlap among ICA subunits.

c. Authors' Conclusions

- The article says that ICA's membership-overlap structure shows closure but not concentration. This means that there are no dominant "hub" subunits, but there is strong triadic closure, which creates a cohesive network with low fragmentation.
- Also, small subunits depend asymmetrically on large ones, meaning that they are more likely to share members, especially with much larger divisions.
- Positive affinity (co-sponsorship) strongly predicts membership overlap, meaning that subunits that co-sponsor conference sessions are more likely to share members. However, this effect remains stable over time and does not increase or decrease.

4. Discussion

a. Implications

- Theoretical: The study connects membership overlap with organizational ecology and network theory. It helps develop a middle-range niche theory, linking the abstract concept of niche overlap to real intra-organizational dynamics within ICA.
- Field structure: The results show high closure, low modularity, and short path lengths, meaning that the ICA network is cohesive rather than fragmented. Subfields are well connected and reachable within the network.
- Policy/practice: The findings show evidence of multiplexity, meaning that subunits can be connected in multiple ways. This suggests that ICA could use co-sponsored conference sessions strategically to increase membership overlap between less connected areas (such as mass communication and interpersonal communication) and promote greater integration in the field.

b. Limitations

- Methodological limitation: TERGMs require binary networks, so the authors had to dichotomize the original weighted membership-overlap data. This means that information about the strength of overlap was lost.
- Generalizability: The study focuses on a single organization (ICA), which limits the generalizability of the findings to other academic associations or organizational contexts.
- Data limitations: The study does not include individual-level socio-demographic data, so the authors cannot precisely distinguish subunits based on the social differences of their members.

c. Future Directions

- Future research could develop and apply valued TERGMs that use the full strength of membership overlap instead of only binary ties. This would allow researchers to test whether the results remain robust when the intensity of overlap between subunits is considered.
- Replicate in other organizations and online communities that allow multi-subunit membership to test whether size dependence and closure generalize
- Extend to post-2015 and post-COVID data and examine how new subunits and disruptions reshape overlap
- Incorporate member socio-demographics to map full niche positions and test who (center x periphery members) is most likely to hold multiple affiliations

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Response Letter (~600 words)

- Your assessment of the study

Overall, the study provides an important contribution to understanding the internal structure of the communication field within the International Communication Association (ICA). By analyzing membership overlap between divisions and interest groups, the authors propose a way to study how scholarly subfields are connected. Instead of focusing on citation or co-authorship networks, the study examines shared memberships, which offers insight into how scholars participate in multiple areas of the field. This also provides insight into how communication research is distributed across different areas, since the field connects with multiple subfields, showing gaps and future research possibilities.

A major difference of the study is the use of network analysis, specifically Temporal Exponential Random Graph Models (TERGMs). This method allows the authors to analyze how relationships between subunits change over time rather than examining a single static network. Using longitudinal data across several years makes it possible to identify structural processes and collaboration patterns that influence the formation of membership overlap ties.

Another important contribution of the study, which I found to be a new perspective, is the integration of organizational ecology and network perspectives. The authors connect theoretical concepts as niche overlap with observable patterns in the network. This helps translate abstract ideas into measurable relationships within an academic organization and contributes to the development of a middle-range niche theory.

- Weaknesses

Despite these strengths, the study also has several limitations, as mentioned above. One methodological limitation is that the analysis requires binary networks, which means that the original membership-overlap data had to be dichotomized. As a result, the model only captures whether overlap exists or not, rather than the strength of the overlap between subunits. This simplification may lead to a loss of information.

Other limitations are that the study focuses only on ICA, which limits its generalizability. In addition, although the article was published in 2024, the data only goes up to 2015, which may limit the understanding of more recent changes in the field. Another limitation is that the study does not include individual-level socio-demographic information about members, which prevents the authors from analyzing differences between members in more detail.

From my perspective, another weakness is that the article use many specific terms and acronyms without clear explanations, which makes the reading process slower. When readers are not familiar with these concepts, they may need to stop frequently to search for definitions while reading. In my opinion, research should be accessible to a broader audience, especially when we consider the connection between academic research and industry or other stakeholders. I believe that this level of technical language may limit how much research can reach people who are not part of the academic environment.

- Other notes

One aspect I noticed in the method is that many measures are not traditional variables, they are statistics. For example, triangle and gwdegree capture structural configurations of the network. The triangle term represents triadic closure, meaning that if two subunits share members with the same third subunit, they are more likely to also share members with each other. The gwdegree measures how connections are distributed across nodes and helps test whether the network becomes concentrated around highly connected subunits. I wonder whether this is a characteristic of this method or if it was simply a choice made by the authors.